

Supplementary Analysis Appendix

Parallel elevation filtering of Ericaceae root-associated fungal communities in Andean páramo ecosystems

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Overview and reproducibility

This document compiles methodological details, supplementary analyses, and reproducibility outputs supporting the main manuscript. Computationally intensive steps (e.g. phylogenetic tree inference, genus-level GLMM fitting, iNEXT3D diversity estimation) were executed on a remote Ubuntu-based computing environment. Their outputs (figures, tables, and selected model objects) are integrated here as static results.

All figures (PNG) and result tables (CSV) are stored within the project repository using relative paths. Rendering this document therefore requires cloning the repository and opening the .qmd file from the repository root.

Data availability.

Repository: [Root_fungi_DADA2](#)

Sequence data: ENA accession number PRJEB107725

Metadata: `Data_S1_metadata.csv`, `Data_S3_soil.csv`

If viewing the PDF version of this Appendix material, note that the HTML version has all code to reproduce these results accessible through drop-downs.

Please visit the project webpage for this: https://ssangulo.github.io/Paramo_SkyIslands_RootFungi/appendix.html

Section 0: Setup

Output directory paths and shared aesthetic constants used throughout this document. All packages are loaded locally within each code section.

Section 1: Data and object loading

All pre-computed R objects are loaded here. Computationally intensive analyses (genus-level GLMMs, iN-EXT3D, NTI/NRI, phylogeny inference) were executed on a remote server; their outputs are stored in `objects/` and loaded as needed.

Section 2: Bioinformatics summary and study design

2.1 Study sites and sampling

This study was conducted across four humid páramo regions in the Colombian Andes, spanning the Central and Eastern Cordilleras. Sampling locations covered elevations from 2,715 to 3,828 m a.s.l., representing the ecological transition from Andean forest to páramo (see main text, Fig. 1). The full sample metadata table is provided in `Data_S1_metadata.csv`.

Root sampling followed standardized sterile procedures. Fine roots of *Gaultheria myrsinoides* were excavated using a spade, and 3–5 cm root tips were clipped using a sterilized pruner. Tools were disinfected with 75% ethanol between samples, and gloves were changed between individual plants to minimize contamination. Root fragments were placed into sealed polybags stored in ice-cooled containers in the field. Samples were processed at Universidad ICESI, where roots were rinsed in sterile water to remove debris, dried briefly under a laminar flow hood, and stored at -20°C prior to shipment to Norway, and long-term storage at -80°C . Soil samples (top 15 cm) were collected at each sampling location, and five aliquots of rinse water were retained as laboratory and transport controls.

Table S1: GPS coordinates, elevation, and vegetation type classification of the 12 sampling locations across four páramo regions. These sites span the forest–subpáramo–páramo gradient and were used for root and soil sampling. In each location, 10 root samples were collected from 5 individuals (2 root samples per individual) and 1 soil sample.

Site ID	Site	Altitude (m)	Latitude (N)	Longitude (W)	Ecosystem
NV_1	Páramo La Nevera	3828	03°30'58.1"	076°03'05.4"	Páramo
NV_2	Páramo La Nevera	3550	03°31'37.7"	076°03'38.7"	Subpáramo
NV_3	Páramo La Nevera	3333	03°32'07.0"	076°04'18.3"	Andean forest
NV_4	Páramo La Nevera	3078	03°32'48.9"	076°04'31.3"	Andean forest
DOM_1	Páramo Las Domínguez	3815	03°43'51.8"	076°06'38.1"	Páramo

DOM_2	Páramo Las Domínguez	3534	03°44'07.2"	076°05'53.2"	Subpáramo
DOM_3	Páramo Las Domínguez	3234	03°44'17.1"	076°05'32.9"	Andean forest
BEL_1	Páramo Belmira	3254	06°38'43.8"	075°40'13.3"	Subpáramo
BEL_2	Páramo Belmira	2950	06°38'22.7"	075°39'57.3"	Andean forest
BEL_3	Páramo Belmira	2715	06°36'55.3"	075°39'51.1"	Andean forest
MA_1	Páramo Matarredonda	3678	04°33'31.5"	074°01'43.8"	Páramo
MA_2	Páramo Matarredonda	3381	04°33'10.3"	073°59'35.3"	Subpáramo

2.2 Read filtering, OTU table, and taxonomic assignment

This section provides supplementary details on sequencing read processing and quality filtering. The full bioinformatic pipeline (including DADA2 processing and replicate filtering) is available in the repository at “/Scripts/full_pipeline_DADA2.R”. Here we report summary tables of read retention and replicate-filtering thresholds used for downstream analyses.

Table S2: Median read numbers per PCR replicate for environmental samples (root and soil samples) and controls (blank extractions, positive, PCR blank, and tag-jump controls). The read counts are grouped at each stage of the DADA2 pipeline. Blank extraction controls (not containing any biological sample) were performed alongside real sample DNA extractions per each extraction batch. PCR blank controls contained all PCR reagents but no DNA template. Tag-jump controls are PCR reactions containing no primers and no DNA template, used to detect tag-jumping (index hopping).

	Raw reads	Filtered	Merged - Non pooled	Merged - Pooled	Merged - Pseudo-pooled
Root samples	77,047	71,783	71,084	70,343	71,242
Soil samples	72,158	66,555	65,311	65,498	65,385
Blank	2,844	2,652	2,618	2,566	2,625
Extraction Controls					
Positive	95,238	88,174	88,062	87,996	87,972
Controls					
PCR Blank	291	261	250	257	256
Controls					
Tag-jump	41	36	27	30	32
Controls					

Table 1: Table S3. Number of OTUs and samples after applying different replicate control filtering thresholds. The table shows results for both soil and root samples combined and only root samples.

Filtering_level	OTU_count_soil+root	Samples_soil+root	OTU_count_root_only	Samples_root_only
No filtering	18451	132	13225	120
OTU present in 2/4	7174	132	4386	120

Filtering_level	OTU_count_soil+root	Samples_soil+root	OTU_count_root_only	Samples_root_only
OTU present in 3/4	4870	131	2966	120
OTU present in 4/4	2875	113	1904	103

Table S4: Summary of the percentage of reads and OTUS with confident taxonomic assignment at each taxonomic rank, represented by habitat. Percentages were calculated from the filtered root-only dataset (soil reads removed).

Rank	Habitat	%_reads_assigned	%_otus_assigned
Phylum	Forest	88,1	88,1
Phylum	Subpáramo	91,6	88,6
Phylum	Páramo	92,9	89,3
Class	Forest	78,3	72,5
Class	Subpáramo	84,2	72,3
Class	Páramo	84,3	76,0
Order	Forest	68,3	59,2
Order	Subpáramo	74,2	57,1
Order	Páramo	78,2	60,9
Genus	Forest	32,8	34,2
Genus	Subpáramo	32,1	32,2
Genus	Páramo	31,4	32,7

2.3 Soil background subtraction

Soil controls were used as a third tier of negative control, alongside PCR blanks and extraction blanks, and were applied as a *count subtraction* rather than as a sample-level exclusion. Within each sampling location, the maximum read count observed per OTU across that location’s co-sampled soil control(s) was subtracted from every root PCR replicate collected at the same location, and any resulting negative value was set to zero. The subtraction was applied to raw read counts, before the two-PCR-replicate detection filter and before PCR replicates were collapsed into samples; soil rows were then removed, leaving the root-only dataset used in all downstream analyses. If a location group contained no matching soil row the per-OTU maximum is undefined, and the affected values were reset to zero. The identical operation had already been applied one step earlier using PCR blanks and extraction blanks as the reference rows.

What this removes, and what it does not. Because the operation subtracts a background level rather than deleting shared taxa, it does not exclude fungi that occur in both soil and roots. An OTU is lost from a given root sample only where its root read count does not exceed the soil maximum at that location; taxa genuinely enriched in roots retain their excess signal. Consistent with this, 1,269 of the 3,689 OTUs recovered from soils were also retained in the final root dataset, and *Leohumicola* — abundant in both compartments — remains one of the dominant root genera (main-text Figure 4C). Root-associated fungi with a soil interface, and dormant propagules of root-associated taxa residing in the soil pool, are therefore represented in the analysed data. The rationale is not that soil-derived taxa are uninformative, but that each root sample is corrected against soil from its own location, which removes precisely the component of root community variation that tracks local soil composition; habitat signal that survives this correction is consequently not attributable to soil carry-over from incomplete root washing or recalcitrant surface-bound cells.

Depth dependence — an acknowledged limitation. The subtraction operates on raw counts and is not normalised for the sequencing depth of the soil control relative to each root sample. Median depths were comparable between the two sample types (65,311 versus 71,084 reads per PCR replicate after merging under the non-pooled strategy; Table S2), so the correction is not grossly asymmetric, but it remains depth-

dependent: a fixed count is removed from root samples that differ in depth, so its proportional effect is larger in shallow-sequenced samples and largest for taxa spanning the soil–root continuum. The procedure therefore trades one source of bias (incomplete root washing, surface-bound soil taxa) for another (depth-dependent magnitude of count removal), and it is conservative with respect to root–soil overlap: genuine overlap is removed together with carry-over. The communities analysed here are accordingly best interpreted as the root-enriched fraction of the fungal community rather than as the complete root mycobiome.

Section 3: Taxonomic overview

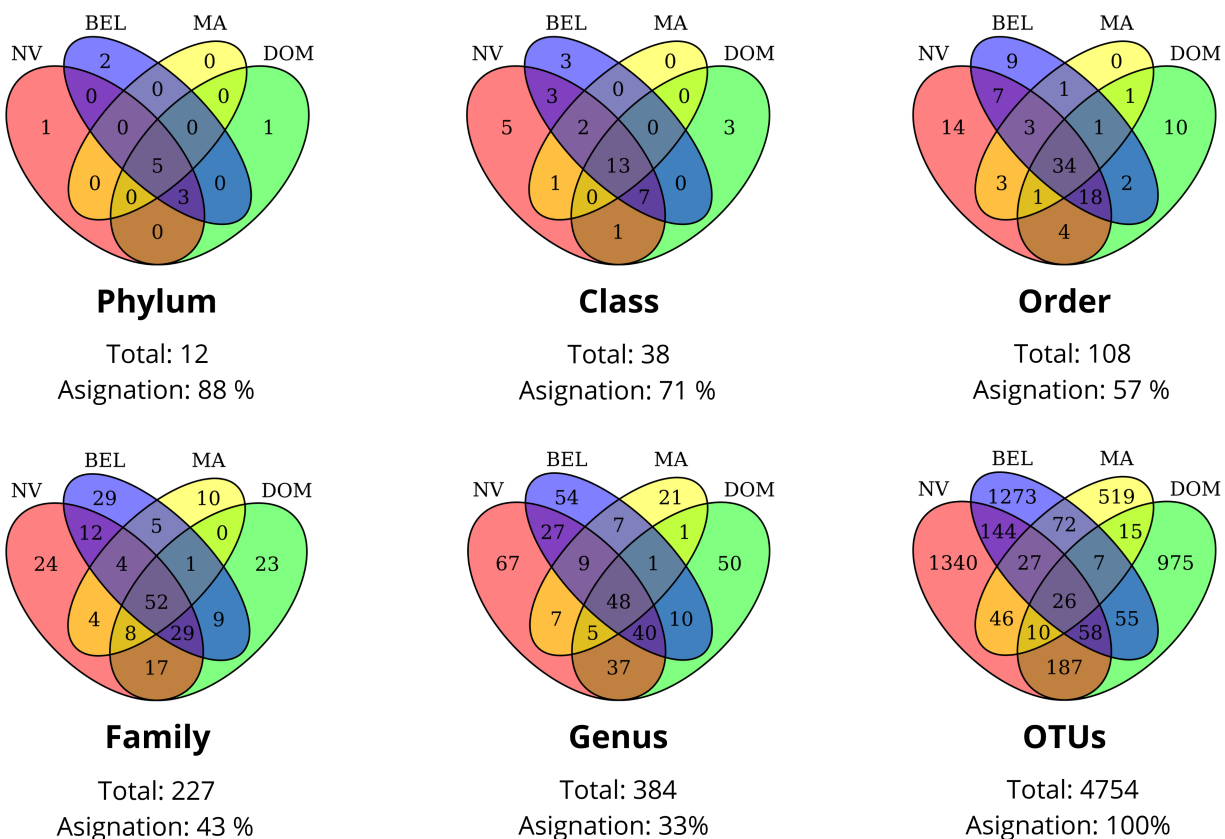


Figure S1: Venn diagram of taxonomic assignments across sampling sites, in the root only dataset, but previous to soil sequences removal. Venn diagrams illustrate the overlap of OTU taxonomic assignments across four study sites: La Nevera (NV), Las Domínguez (DOM), Belmira (BEL), and Matarredonda (MA). Only OTUs with confirmed assignments at each taxonomic level are included.

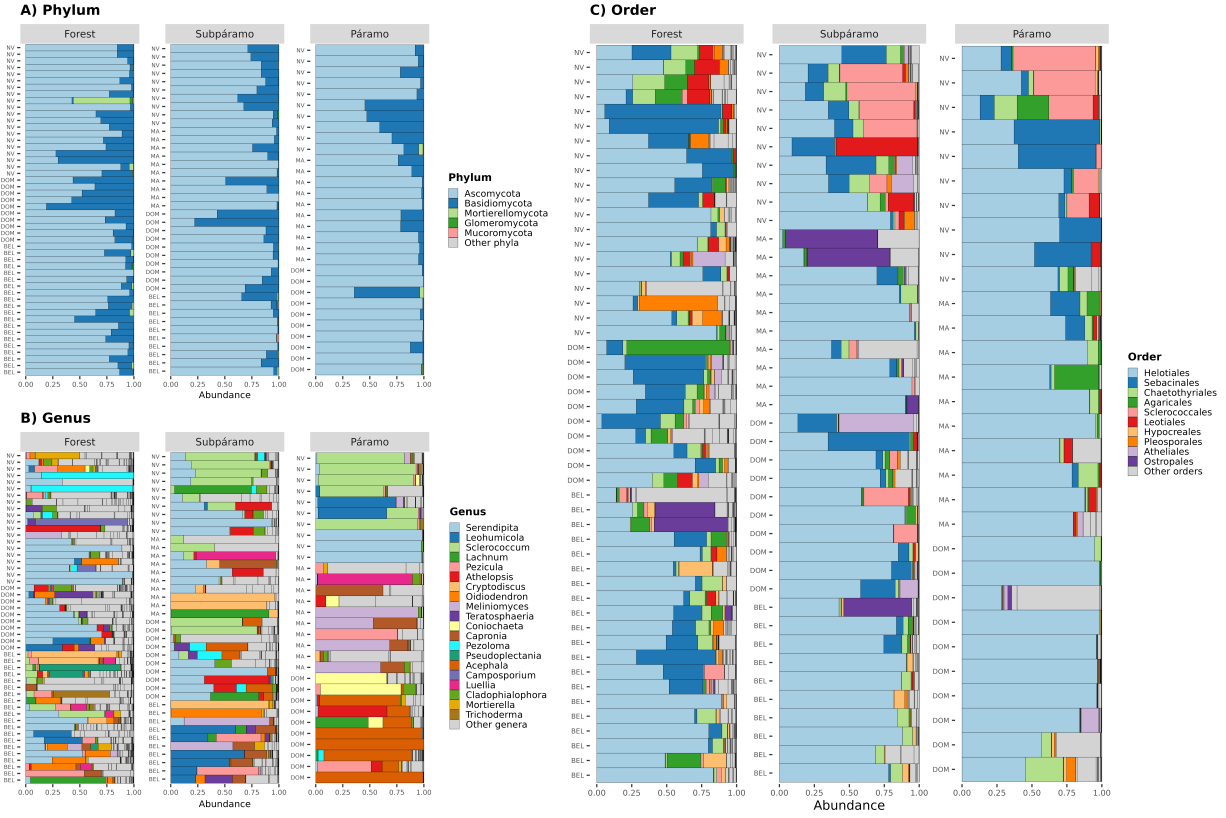


Figure S2: Taxonomic composition in root samples along the elevational gradient. Relative read abundances are shown at three taxonomic ranks: A) Phylum, B) Genus and C) Order. Each bar represents an individual root sample, grouped by habitat: Forest (2700-3300 m), Subpáramo (3300-3500 m), and Páramo (3800 m). Study sites are indicated on the y-axis.

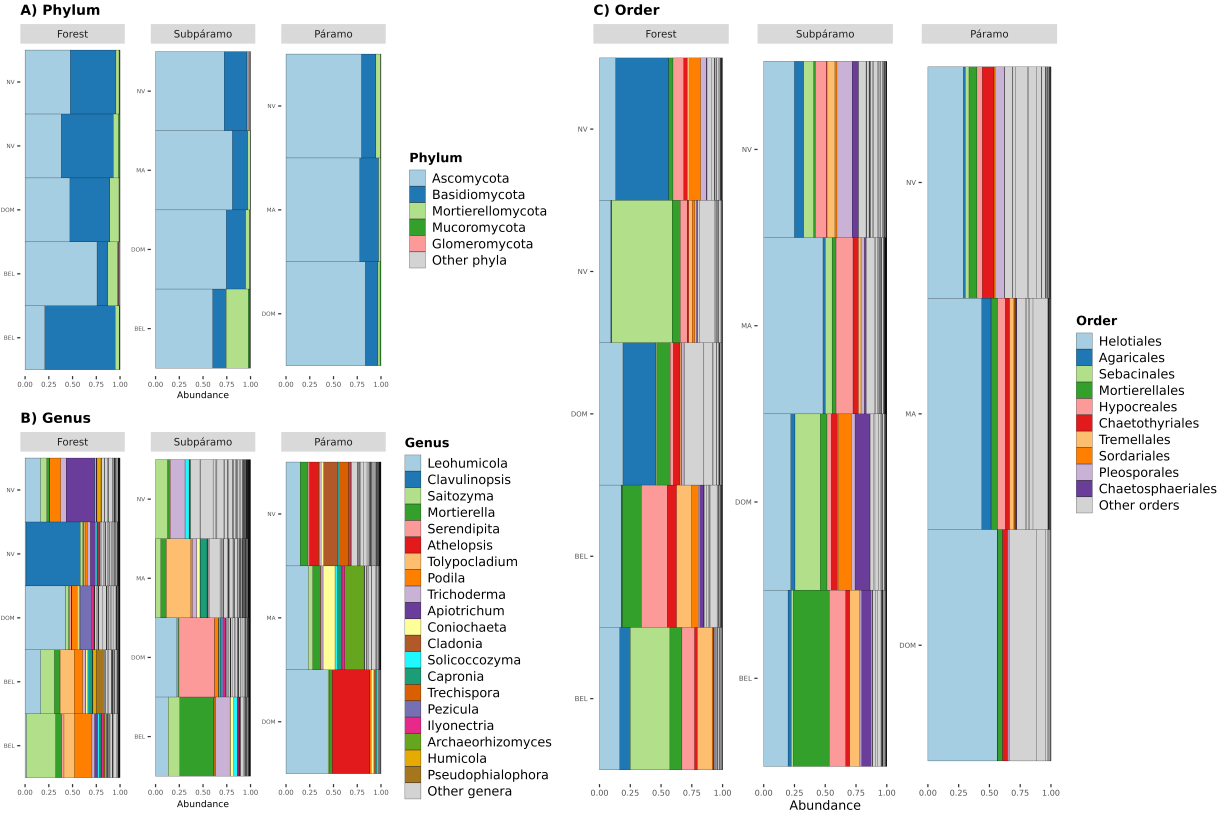


Figure S3: Taxonomic composition in soil samples, collected at each of the sampling locations. Relative read abundances are shown at three taxonomic ranks: A) Phylum, B) Genus, and C) Order. Each bar represents an individual soil sample, grouped by habitat: Forest (2700-3300 m), Subpáramo (3300-3500 m), and Páramo (3800 m). Study sites are indicated on the y-axis.

Section 4: Community composition

This section provides detailed statistical analyses supporting the patterns of root-associated fungal community composition described in the main text. Several of these analyses form the basis of results reported in the main manuscript; here we document the full model specifications and test statistics to ensure transparency and reproducibility.

We include distance-based redundancy analysis (dbRDA) based on robust Aitchison distances to assess habitat-associated compositional structure while conditioning on site. We further present PERMANOVA results from a balanced subset of samples (NV-DOM) used to test habitat and site effects under a controlled sampling design. In addition, we report a centroid-based turnover model quantifying the magnitude of community change across habitats using the full dataset.

4.1 dbRDA ordination (Tables S5–S6)

Distance-based redundancy analysis (dbRDA) was conducted on robust Aitchison distances to test for habitat-associated compositional structure while conditioning on site. This analysis is reported in the main manuscript as Figure 2A (habitat separation conditioned on site; permutation test $F = 1.41$, $p = 0.001$).

Table 2: Table S5. dbRDA permutation tests (by term; 999 permutations).

Component	Df	Variance	F	Pr
habitat	2	22.718	1.408	0.001
Residual	54	435.501	NA	NA

Table 3: Table S6. dbRDA permutation tests (by constrained axis; 999 permutations).

Axis	Df	Variance	F	Pr
CAP1	1	14.579	1.808	0.001
CAP2	1	8.139	1.009	0.470
Residual	54	435.501	NA	NA

4.2 PERMANOVA and beta-dispersion (Tables S7–S9)

Beta-dispersion tests on the balanced NV–DOM subset indicated higher within-habitat variability in forests (mean distance to centroid 23.6) compared with subpáramo (17.8) and páramo (17.1) communities ($F = 11.30$, $p = 0.001$), whereas dispersion did not differ among sites ($p = 0.571$). Part of the PERMANOVA habitat effect may therefore reflect differences in dispersion rather than location. The significant habitat $\times$ site interaction ($R^2 = 0.09$, $p = 0.001$) indicates that the habitat effect on composition is not identical between the two complete gradients: a shared habitat response with a site-specific component.

Table 4: Table S7. PERMANOVA (adonis2) results by term.

Source	df	SumOfSqs	R2	Pseudo.F	Pr..F.
site	1	742.078	0.056	1.786	0.001
habitat	2	1351.164	0.102	1.626	0.001
site:habitat	2	1241.269	0.093	1.493	0.001
Residual	24	9974.546	0.749	NA	NA
Total	29	13309.057	1.000	NA	NA

Table 5: Table S8. Habitat beta-dispersion (distance to centroid). Permutation test $p = 0.001$.

habitat	avg_distance_to_centroid
forest	23.631
paramo	17.147
subparamo	17.794

Table 6: Table S9. Site beta-dispersion permutation test.

test	permutations	F	p_value
betadisper_site	999	0.314	0.571

4.3 Centroid-based turnover model (Table S10)

To quantify the magnitude of community turnover across habitats while accounting for spatial structure, we calculated the Aitchison distance of each sample to its site-specific centroid and modelled turnover as a func-

tion of habitat, site, and their interaction. Table S10 reports sequential (Type I) tests, in which habitat is not adjusted for site. Adjusted for site, habitat still affected turnover magnitude ($F_{2,54} = 6.11$, $p = 0.004$), with forest plants further from their site centroid than subpáramo (difference 3.6, $p = 0.008$) or páramo plants (3.9, $p = 0.015$), whereas the habitat $\times$ site interaction was not significant, indicating consistent habitat-associated turnover magnitude across geographically isolated páramo regions (`adjusted_tests_centroid_nri.csv`).

Panel B in the main manuscript Figure 2 summarizes this model (habitat adjusted for site $F_{2,54} = 6.11$, $p = 0.004$; habitat $\times$ site $F_{4,50} = 1.28$, $p = 0.29$), with Matarredonda (MA) shown as a dashed trend due to a compressed gradient.

Table 7: Table S10. Linear model results for centroid-based turnover.

Effect	df	Sum.of.Squares	Mean.Square	F.value	p.value
habitat	2	376.745	188.373	14.719	0.000
site	3	138.096	46.032	3.597	0.020
habitat:site	4	65.496	16.374	1.279	0.291
Residuals	50	639.914	12.798	NA	NA

4.4 Figure 2

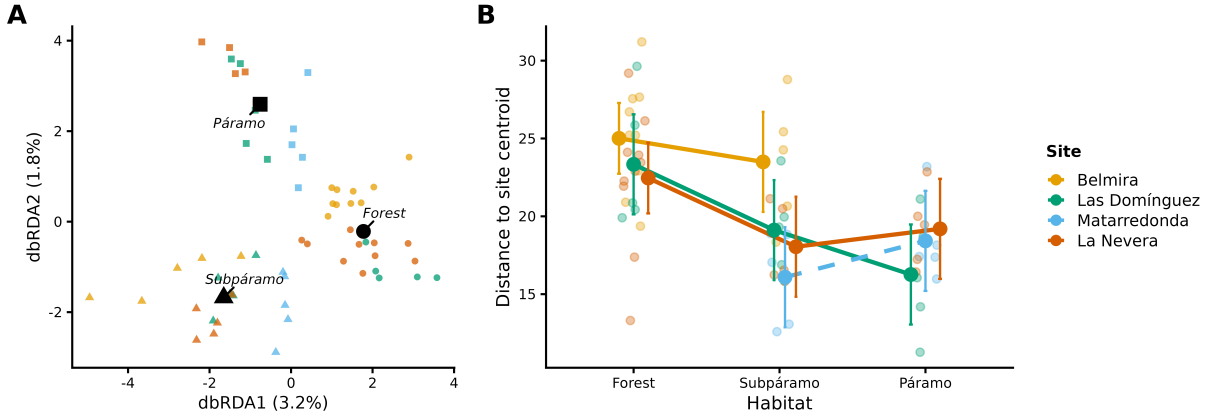


Figure 2: Community composition and turnover across habitats. Panel A: dbRDA ordination of RAF communities conditioned on site (color = site, shape = habitat; permutation $F = 1.41$, $p = 0.001$). Panel B: centroid-based turnover model (distance to site-specific centroid across habitats), with MA shown as dashed due to incomplete gradient (habitat $F_{2,50} = 14.7$, $p < 0.001$; habitat $\times$ site $F_{4,50} = 1.28$, $p = 0.29$).

Section 5: Diversity trajectories

Diversity trajectories were assessed using three complementary approaches: per-plant Hill diversity models (rarefied to 28,122 reads) to test parallelism across sites, iNEXT3D rarefaction/extrapolation curves for habitat-level estimates with plants as sampling units, and beta-diversity partitioning. All analyses require a rooted phylogenetic tree, inferred below. Analyses are presented in computation order; Figure 3 follows as a composite of pre-computed outputs.

Alignment and tree inference. OTU reference sequences (ITS2; the 4,386 OTUs of the replicate-filtered, soil-subtracted root dataset, retaining both root replicates per plant) were aligned with the profile-to-profile aligner `AlignSeqs` in DECIPHER (Wright 2016), with sequence anchoring disabled (`anchor = NA`). No post-alignment curation, masking or trimming was applied. ITS2 is length-variable and hypervariable, so automated trimming heuristics discard a large share of the phylogenetically informative sites; and because the alignment is used only to derive a single fixed topology on which all samples are subsequently compared,

alignment uncertainty enters every sample’s diversity estimate identically rather than differentially by habitat. A neighbour-joining tree built from maximum-likelihood distances (`dist.ml`) provided the starting topology, which was then optimised by maximum likelihood with `optim.pml` in `phangorn` (Schliep 2011) under GTR + Γ + I with four discrete rate categories, simultaneously optimising base frequencies, the rate matrix, branch lengths, the gamma shape parameter and the proportion of invariant sites, using stochastic rearrangements. The tree was inferred unrooted and midpoint-rooted (`phangorn::midpoint`) before use in phylogenetic diversity, NRI and NTI calculations.

Substitution model. GTR + Γ + I was adopted a priori rather than selected by a formal model-selection test, following the standard DADA2/phyloseq phylogenetic workflow (Callahan et al. 2016). It is the most parameter-rich time-reversible model available in `optim.pml`, so under-parameterisation is not a concern, and a formal model test across 4,386 ITS2 sequences was computationally prohibitive at this scale. All phylogenetic metrics reported here (phylogenetic Hill numbers as meanPD, NRI and NTI) are comparative statistics computed from relative branch lengths on this single shared topology, and are therefore expected to be robust to moderate differences in substitution model.

5.1 Incidence-based beta-diversity partitioning (Table S11)

Sørensen dissimilarities were partitioned into turnover and nestedness-resultant components across taxonomic levels using the `betapart` package, on presence–absence data rarefied to 28,122 reads (the minimum observed library), retaining all 60 per-plant communities.

Table 8: Table S11. Sørensen beta-diversity partitioning across taxonomic levels and sites. SIM = turnover (taxon replacement), SNE = nestedness-resultant component, SOR = total Sørensen dissimilarity (SIM + SNE); N_units = number of taxonomic units resolved at that rank; Turnover_pct = SIM as a percentage of SOR. Computed with `beta.multi()` on presence–absence data rarefied to 28,122 reads.

Site	Taxonomic_Level	N_units	beta.SIM	beta.SNE	beta.SOR	Turnover_pct
DOM	OTU	1158	0.9389	0.0147	0.9536	98.5
DOM	Genus	283	0.8366	0.0399	0.8764	95.5
DOM	Order	88	0.6998	0.0748	0.7746	90.3
DOM	Class	32	0.5562	0.1243	0.6805	81.7
DOM	Phylum	11	0.3737	0.2616	0.6353	58.8
NV	OTU	1607	0.9517	0.0133	0.9650	98.6
NV	Genus	333	0.8535	0.0454	0.8989	94.9
NV	Order	100	0.7333	0.0904	0.8236	89.0
NV	Class	38	0.6405	0.1188	0.7593	84.4
NV	Phylum	10	0.5153	0.1946	0.7099	72.6
BEL	OTU	1405	0.9300	0.0138	0.9438	98.5
BEL	Genus	286	0.8026	0.0464	0.8490	94.5
BEL	Order	96	0.6867	0.0823	0.7690	89.3
BEL	Class	37	0.5869	0.1094	0.6963	84.3
BEL	Phylum	12	0.3985	0.2444	0.6429	62.0
MA	OTU	594	0.9264	0.0131	0.9395	98.6
MA	Genus	145	0.7582	0.0638	0.8221	92.2
MA	Order	59	0.6234	0.0924	0.7159	87.1
MA	Class	21	0.3624	0.1873	0.5498	65.9
MA	Phylum	7	0.1818	0.2513	0.4331	42.0

5.2 iNEXT3D rarefaction curves and diversity summaries (Tables S12–S13b)

We quantified habitat-associated patterns in taxonomic diversity (TD) and phylogenetic diversity (PD) using iNEXT3D and Hill numbers ($q = 0, 1, 2$). Diversity estimates were computed from incidence (presence/absence) matrices at the habitat level, with plants as sampling units (an OTU is present in a plant

if detected in either of its two root samples; 25, 20 and 15 plants in forest, subpáramo and páramo), and evaluated under rarefaction/extrapolation with bootstrap confidence intervals (nboot = 500). Phylogenetic diversity was estimated as meanPD (effective number of lineages) using a pruned phylogeny matched to the observed taxa. The combined curve visualization is presented in main-text Figure 3A; asymptotic summaries are provided in Tables S12–S13, coverage-standardised estimates in Table S13b, and the per-plant parallelism models in Tables S14–S15.

Sample coverage at the reference sample size is 0.58, 0.42 and 0.49 in forest, subpáramo and páramo, and 70–79% of the OTUs of each habitat are detected in a single plant, so the asymptotic estimates of Tables S12–S13 rest on a large undetected fraction and on habitats of unequal completeness. Habitats are therefore compared at a common sample coverage of 60% (Table S13b), the level iNEXT adopts by default: it is the coverage subpáramo attains at twice its observed number of plants, which is the recommended limit for extrapolation. Páramo is roughly half as diverse as forest and subpáramo for every metric and order, with non-overlapping confidence intervals. Forest and subpáramo do not differ in TD at any order (ratios 1.00–1.07, overlapping CIs) and differ only slightly in PD (ratios 1.04–1.21). Habitat-level diversity therefore follows forest > subpáramo > páramo rather than a steady decline.

Table 9: Table S12. Taxonomic diversity (TD) summary from iNEXT3D across habitats with plants as sampling units ($q = 0, 1, 2$; nboot = 500). Observed and asymptotic estimates are reported with standard errors and 95% confidence intervals.

Habitat	qTD	TD_obs	TD_asy	s.e.	qTD.LCL	qTD.UCL	Diversity_order
Forest	Species richness	2521.000	6172.227	198.792	5782.601	6561.852	Species richness
Forest	Shannon diversity	1992.149	4148.349	91.922	3968.186	4328.513	Shannon diversity
Forest	Simpson diversity	1428.114	2101.863	56.685	1990.763	2212.963	Simpson diversity
Páramo	Species richness	901.000	2407.246	130.249	2151.962	2662.529	Species richness
Páramo	Shannon diversity	765.978	1859.858	75.676	1711.536	2008.180	Shannon diversity
Páramo	Simpson diversity	600.227	1051.258	50.412	952.452	1150.064	Simpson diversity
Subpáramo	Species richness	1533.000	4695.445	202.951	4297.668	5093.221	Species richness
Subpáramo	Shannon diversity	1340.805	3825.086	127.254	3575.673	4074.500	Shannon diversity
Subpáramo	Simpson diversity	1066.501	2128.877	93.427	1945.763	2311.990	Simpson diversity

Table 10: Table S13. Phylogenetic diversity (PD; meanPD) summary from iNEXT3D across habitats with plants as sampling units ($q = 0, 1, 2$; nboot = 500). Observed and asymptotic effective numbers of lineages are reported with standard errors and 95% confidence intervals.

Habitat	qPD	PD_obs	PD_asy	s.e.	qPD.LCL	qPD.UCL	Reftime	Type
Forest	$q = 0$ PD	185.553	326.866	10.346	306.589	347.144	0.873	meanPD
Forest	$q = 1$ PD	112.466	148.345	2.022	144.382	152.308	0.873	meanPD
Forest	$q = 2$ PD	70.612	77.842	0.815	76.244	79.439	0.873	meanPD
Páramo	$q = 0$ PD	81.971	143.212	6.652	130.173	156.250	0.873	meanPD
Páramo	$q = 1$ PD	56.100	80.094	1.874	76.420	83.767	0.873	meanPD
Páramo	$q = 2$ PD	37.576	43.701	0.821	42.093	45.310	0.873	meanPD
Subpáramo	$q = 0$ PD	132.936	246.757	10.004	227.149	266.364	0.873	meanPD
Subpáramo	$q = 1$ PD	88.309	130.400	2.673	125.161	135.638	0.873	meanPD
Subpáramo	$q = 2$ PD	54.604	63.591	0.974	61.682	65.500	0.873	meanPD

Table 11: Table S13b. Coverage-standardised taxonomic (TD) and phylogenetic (PD; meanPD) diversity by habitat at 60% sample coverage, with plants as sampling units ($q = 0, 1, 2$; $n_{boot} = 500$). Estimates are reported with bootstrap standard errors and 95% confidence limits.

diversity	q	habitat	estimate	se	lcl	ucl	coverage
TD	0	forest	2710.135	69.099	2574.704	2845.566	0.6
TD	1	forest	2105.809	49.605	2008.585	2203.033	0.6
TD	2	forest	1475.376	34.114	1408.513	1542.239	0.6
TD	0	subparamo	2532.624	102.558	2331.615	2733.634	0.6
TD	1	subparamo	2104.899	80.148	1947.811	2261.987	0.6
TD	2	subparamo	1421.084	53.180	1316.852	1525.316	0.6
TD	0	paramo	1229.433	52.810	1125.928	1332.939	0.6
TD	1	paramo	999.084	38.746	923.143	1075.026	0.6
TD	2	paramo	708.818	25.987	657.883	759.752	0.6
PD	0	forest	195.707	3.520	188.809	202.606	0.6
PD	1	forest	115.492	1.488	112.576	118.409	0.6
PD	2	forest	71.273	0.715	69.872	72.674	0.6
PD	0	subparamo	188.822	4.775	179.463	198.181	0.6
PD	1	subparamo	110.342	2.095	106.236	114.448	0.6
PD	2	subparamo	58.756	0.820	57.149	60.362	0.6
PD	0	paramo	102.308	3.132	96.169	108.447	0.6
PD	1	paramo	64.378	1.441	61.554	67.202	0.6
PD	2	paramo	39.556	0.691	38.201	40.911	0.6

5.3 Per-plant Hill diversity: parallelism models (Tables S14–S15)

To evaluate whether diversity trajectories are parallel across geographically isolated sites, we model diversity as a function of habitat (ordered along elevation) and site, each tested adjusted for the other, and test the habitat $\times$ site interaction by comparing the additive and interaction models, cross-checked against AICc. The interaction test is itself the parallelism result, and it determines which model the habitat effects are then read from: where the interaction is not supported the additive model is retained and habitat is reported marginally, and where it is supported there is no common trajectory and habitat effects are reported within sites. The same rule is applied to all six responses.

For this section, diversity is computed independently for each plant using Hill numbers of taxonomic diversity (TD) and phylogenetic diversity (PD; meanPD) at $q = 0, 1$ and 2 , after rarefying every plant to 28,122 reads (the smallest library) with `iNEXT.3D::estimate3D`. Coverage-based standardisation is not informative here: no plant retains a singleton OTU after the 2-of-4 replicate filter, so coverage is 0.999 at any depth.

Models are gamma GLMs with a log link, so habitat effects are ratios of effective diversity; gamma had the lowest summed AICc in a Jacobian-corrected screen against Gaussian and log-normal models, and habitat results were refitted under all three families as a sensitivity check.

Adjusted for site, habitat affects per-plant richness (TD $q = 0$: $F_{2,54} = 9.21$, $p = 0.0004$; PD $q = 0$: $F_{2,54} = 9.61$, $p = 0.0003$), with the pairwise contrasts reported below (Table S15b). The habitat effect is still present at $q = 1$ (TD $F_{2,54} = 3.21$, $p = 0.048$), though AICc marginally prefers dropping habitat at this order and no pairwise contrast resolves there, and it is gone by $q = 2$ (TD $p = 0.15$). The habitat $\times$ site interaction is absent for TD at every order ($p = 0.22$ – 0.25) and for PD at $q = 0$ ($p = 0.42$), but present for abundance-weighted PD ($q = 1$: $F_{4,50} = 3.81$, $p = 0.009$; $q = 2$: $F_{4,50} = 7.27$, $p = 0.0001$), where it is driven by the single forest location at Las Domínguez. AICc agrees with every one of these decisions: the additive model carries Akaike weights of 0.91 (TD $q = 0$) and 0.96 (PD $q = 0$) against the interaction model, while at PD $q = 1$ and $q = 2$ the interaction model carries 0.89 and 1.00.

Post-hoc pairwise habitat contrasts (Table S15b). Habitat is a categorical predictor, so the omnibus F -test for the habitat term establishes that habitats differ but does not identify which pairs differ, or in

which direction. We therefore followed it with pairwise contrasts among the three habitats, taken from the same additive model as the omnibus test (gamma GLM with a log link, $\text{diversity} \sim \text{habitat} + \text{site}$) and Tukey-adjusted for the three comparisons. Taking both tests from one model matters: in the published version the omnibus test was sequential (Type I, habitat entered before site and therefore not adjusted for it) while the contrasts came from the additive model, so the two asked different questions and a significant habitat term could sit alongside no resolved pair. The full design also contains empty habitat $\times$ site cells (Matarredonda has no forest, Belmira no páramo), which makes habitat marginal means non-estimable from the interaction model. Contrasts are reported on the response scale as ratios of effective diversity, so a ratio above one indicates higher diversity in the first habitat.

The per-plant decline is concentrated at the forest boundary. At $q = 0$ both forest contrasts are resolved for each metric (TD forest/subpáramo 1.52, $p = 0.0005$; forest/páramo 1.69, $p = 0.0002$; PD 1.39, $p = 0.001$ and 1.58, $p = 0.0001$), while subpáramo and páramo do not differ (TD 1.11, $p = 0.64$; PD 1.13, $p = 0.40$): marginal richness falls from forest (TD 146.9, PD 20.1) to subpáramo (96.6, 14.4) and páramo (87.1, 12.7), with the second step flat. At $q = 1$ the direction is the same but nothing resolves (TD forest/subpáramo 1.48, $p = 0.055$; forest/páramo 1.53, $p = 0.078$), and at $q = 2$ no TD contrast is resolved at all. Polynomial contrasts across the ordered habitats are significant for TD at $q = 0$ ($p < 0.0001$) and $q = 1$ ($p = 0.031$) and for PD at $q = 0$ ($p = 0.00004$), so diversity does fall along the gradient — but as one step at the treeline rather than a stepwise decline across successive habitats.

For abundance-weighted PD ($q = 1-2$) the habitat $\times$ site interaction is supported (Table S15), so no common habitat trajectory applies and neither the marginal contrasts nor the polynomial trends of the additive model are inferential there; the corresponding rows of Table S15b are reported for completeness only. From the interaction model, the habitat effect at these orders is confined to Las Domínguez, where forest plants exceed subpáramo ($q = 1$: $1.69\times$, $p = 0.002$; $q = 2$: $1.62\times$, $p < 0.001$) and páramo ($1.81\times$, $p = 0.0004$; $1.55\times$, $p < 0.001$), while no habitat contrast is resolved at any other site. Per-habitat sample sizes for pairwise inference remain small ($n = 25, 20$ and 15 plants for forest, subpáramo and páramo).

This differs from the habitat-level pattern in Section 5.2, where forest and subpáramo are indistinguishable and páramo is clearly the poorest assemblage: per plant the break is at the forest boundary, per habitat it is at the páramo boundary. The two are consistent if subpáramo plants are individually poorer than forest plants but more different from one another, so that the habitat total keeps pace with forest, while páramo plants are individually as rich as subpáramo plants but more alike. This comparison is qualitative only, since the per-plant values are abundance-based and the habitat-level values incidence-based.

Table 12: Table S14. Tests for per-plant taxonomic diversity (TD; Hill $q = 0, 1, 2$; rarefied to 28,122 reads) from gamma GLMs (log link): habitat and site each adjusted for the other (additive model), and the habitat $\times$ site interaction by model comparison. The interaction is not significant at any order ($p = 0.22-0.25$), supporting parallel TD trajectories across sites.

diversity	q	term	df1	df2	F	p
TD	0	habitat x site (parallelism)	4	50	1.4977	0.2171
TD	0	habitat (adjusted for site)	2	54	9.2123	0.0004
TD	0	site (adjusted for habitat)	3	54	0.8080	0.4950
TD	1	habitat x site (parallelism)	4	50	1.4319	0.2372
TD	1	habitat (adjusted for site)	2	54	3.2076	0.0483
TD	1	site (adjusted for habitat)	3	54	0.4513	0.7175
TD	2	habitat x site (parallelism)	4	50	1.4023	0.2468
TD	2	habitat (adjusted for site)	2	54	1.9999	0.1453
TD	2	site (adjusted for habitat)	3	54	0.2829	0.8375

Table 13: Table S15. Tests for per-plant phylogenetic diversity (PD; meanPD; Hill $q = 0, 1, 2$; rarefied to 28,122 reads) from gamma GLMs (log link): habitat and site each adjusted for the other (additive model), and the habitat $\times$ site interaction by model comparison. The interaction is not significant at $q = 0$ ($p = 0.42$) but is at $q = 1$ ($p = 0.009$) and $q = 2$ ($p = 0.0001$), driven by Las Domínguez.

diversity	q	term	df1	df2	F	p
PD	0	habitat x site (parallelism)	4	50	1.0012	0.4158
PD	0	habitat (adjusted for site)	2	54	9.6132	0.0003
PD	0	site (adjusted for habitat)	3	54	0.6139	0.6089
PD	1	habitat x site (parallelism)	4	50	3.8086	0.0089
PD	1	habitat (adjusted for site)	2	54	4.0743	0.0225
PD	1	site (adjusted for habitat)	3	54	3.3884	0.0244
PD	2	habitat x site (parallelism)	4	50	7.2732	0.0001
PD	2	habitat (adjusted for site)	2	54	4.5366	0.0151
PD	2	site (adjusted for habitat)	3	54	6.5005	0.0008

Table 14: Table S15b. Post-hoc pairwise contrasts among habitats for per-plant Hill diversity ($q = 0, 1, 2$; rarefied to 28,122 reads), for taxonomic (TD) and phylogenetic (PD; meanPD) diversity. Contrasts come from the same additive model as the omnibus tests in Tables S14–S15 (gamma GLM with a log link, diversity $\sim$ habitat + site) and are Tukey-adjusted for three comparisons; the additive model is used because the full design contains empty habitat $\times$ site cells, which makes habitat marginal means non-estimable from the interaction model. Estimates are on the response scale: ratio is the multiplicative difference in effective diversity between the two habitats (ratio > 1 = higher diversity in the first habitat), with 95% confidence limits. For PD at $q = 1$ and $q = 2$ the habitat $\times$ site interaction is supported (Table S15), so the additive model does not describe those orders and their rows are descriptive rather than inferential.

diversity	q	contrast	ratio	SE	df	lower.CL	upper.CL	t.ratio	p.value
TD	0	forest / subparamo	1.5204	0.1590	54	1.1817	1.9562	4.0065	0.0005
TD	0	forest / paramo	1.6864	0.2047	54	1.2586	2.2595	4.3045	0.0002
TD	0	subparamo / paramo	1.1092	0.1280	54	0.8399	1.4647	0.8980	0.6439
TD	1	forest / subparamo	1.4781	0.2438	54	0.9933	2.1996	2.3694	0.0549
TD	1	forest / paramo	1.5270	0.2924	54	0.9626	2.4225	2.2108	0.0784
TD	1	subparamo / paramo	1.0331	0.1880	54	0.6663	1.6018	0.1788	0.9825
TD	2	forest / subparamo	1.4229	0.2573	54	0.9203	2.2001	1.9506	0.1346
TD	2	forest / paramo	1.3258	0.2783	54	0.7994	2.1989	1.3435	0.3777
TD	2	subparamo / paramo	0.9318	0.1859	54	0.5761	1.5070	-0.3542	0.9333
PD	0	forest / subparamo	1.3936	0.1220	54	1.1285	1.7210	3.7907	0.0011
PD	0	forest / paramo	1.5813	0.1607	54	1.2377	2.0202	4.5084	0.0001
PD	0	subparamo / paramo	1.1347	0.1096	54	0.8990	1.4321	1.3082	0.3969
PD	1	forest / subparamo	1.1801	0.0982	54	0.9656	1.4422	1.9897	0.1245
PD	1	forest / paramo	1.2653	0.1223	54	1.0024	1.5970	2.4348	0.0472
PD	1	subparamo / paramo	1.0721	0.0985	54	0.8593	1.3377	0.7585	0.7298

diversity	q	contrast	ratio	SE	df	lower.CL	upper.CL	t.ratio	p.value
PD	2	forest / subparamo	1.1620	0.0639	54	1.0179	1.3266	2.7329	0.0227
PD	2	forest / paramo	1.1453	0.0731	54	0.9821	1.3356	2.1263	0.0940
PD	2	subparamo / paramo	0.9856	0.0598	54	0.8516	1.1406	-0.2396	0.9689

5.4 Figure 3: Diversity trajectories

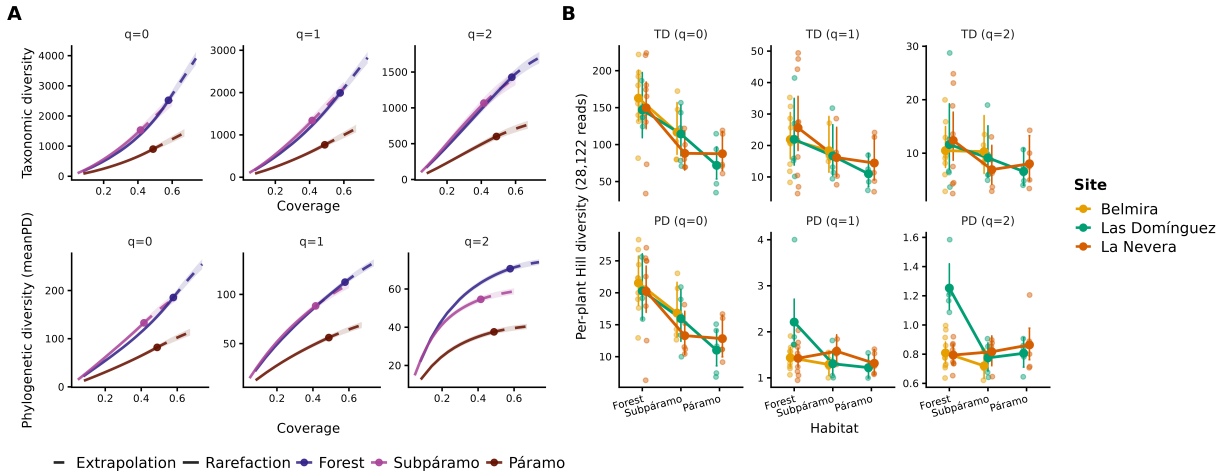


Figure 3: Diversity across the elevational gradient. Panel A shows iNEXT3D coverage-based rarefaction/extrapolation curves for TD and PD (meanPD) at $q = 0, 1, 2$ by habitat, with plants as sampling units; páramo is lower than forest and subpáramo, which overlap. Panel B shows per-plant Hill diversity (rarefied to 28,122 reads) at $q = 0, 1, 2$ for TD and PD, with model-estimated means $\pm$ 95% CI for BEL, DOM, and NV (MA excluded). Habitat $\times$ site interactions were non-significant for TD at every order and for PD at $q = 0$, but significant for PD at $q = 1$ ($F_{4,50} = 3.81$, $p = 0.009$) and $q = 2$ ($F_{4,50} = 7.27$, $p = 0.0001$), driven by Las Domínguez.

Section 6: Lineage-specific responses

6.1 Beta regression models (Tables S16–S18)

To quantify elevational trends in the relative abundance of dominant fungal lineages, we fitted beta-regression models to order-level relative read abundances. Models were fitted separately for each order using a logit link, with elevation as a continuous predictor and site included as a fixed effect. Relative abundances were calculated as proportions of total reads per plant ($n = 60$) and transformed using a Smithson–Verkuilen adjustment.

We focus on helotiales and sebacinales, the two most abundant orders, which also showed the only significant elevation responses. For both orders, fitted relationships and raw data are shown in Figure 4 (Panels A-B), while full model summaries are provided in Tables S16 and S17. Elevation effects for the eight most abundant orders are summarized in Table S18.

Coefficients are on the logit scale: the Helotiales elevation coefficient ($+0.101$ per 100 m, $p = 0.017$) corresponds to $+10.6\%$ in the odds, or about $+2.5$ percentage points per 100 m at the mean Helotiales proportion (0.43); the Sebacinales coefficient (-0.100 per 100 m, $p = 0.033$) corresponds to about -0.7 percentage points. Site-specific elevation slopes were not supported for either order (elevation $\times$ site likelihood-ratio test: Helotiales $p = 0.060$, Sebacinales $p = 0.33$; ΔAICc $+0.6$ and $+4.6$), so the additive model is retained.

Model diagnostics were examined to assess fit and identify potential violations of distributional assumptions. Diagnostic plots include Pearson residuals versus fitted values and elevation, quantile residual Q-Q plots, and leverage values (Figures S4–S5).

Table 15: Table S16. Beta regression summary for Helotiales (mean model coefficients and precision (phi)).

Model	Term	Estimate	SE	z	p
helotiales	(Intercept)	-3.2040	1.2559	-2.5512	0.0107
helotiales	elevation	0.0010	0.0004	2.3959	0.0166
helotiales	siteDOM	-0.7291	0.3651	-1.9969	0.0458
helotiales	siteMA	0.0050	0.3871	0.0130	0.9896
helotiales	siteNV	-0.8864	0.3346	-2.6494	0.0081
helotiales	(phi) (phi)	6.2698	1.0711	5.8534	0.0000

Table 16: Table S17. Beta regression summary for Sebacinales (mean model coefficients and precision (phi)).

Model	Term	Estimate	SE	z	p
sebacinales	(Intercept)	0.4169	1.3941	0.2990	0.7649
sebacinales	elevation	-0.0010	0.0005	-2.1320	0.0330
sebacinales	siteDOM	0.7196	0.4162	1.7289	0.0838
sebacinales	siteMA	0.3029	0.4603	0.6580	0.5105
sebacinales	siteNV	0.9779	0.3714	2.6333	0.0085
sebacinales	(phi) (phi)	10.1810	2.0241	5.0298	0.0000

Table 17: Table S18. Elevation effects from beta regression models for the dominant orders (mean submodel), with pseudo- R^2 .

order	term	estimate	se	z	p_value	pseudo_R2
helotiales	elevation	1e-03	4e-04	2.3959	0.0166	0.2133
sebacinales	elevation	-1e-03	5e-04	-2.1320	0.0330	0.2261
chaetothyriales	elevation	1e-04	4e-04	0.2327	0.8160	0.0239
sclerococcales	elevation	7e-04	5e-04	1.3758	0.1689	0.1330
agaricales	elevation	-7e-04	4e-04	-1.7597	0.0785	0.0929
pleosporales	elevation	-4e-04	4e-04	-1.0668	0.2861	0.1528
hypocreales	elevation	-2e-04	3e-04	-0.7300	0.4654	0.2397
leotiales	elevation	-2e-04	3e-04	-0.6812	0.4957	0.2964

Helotiales model diagnostics

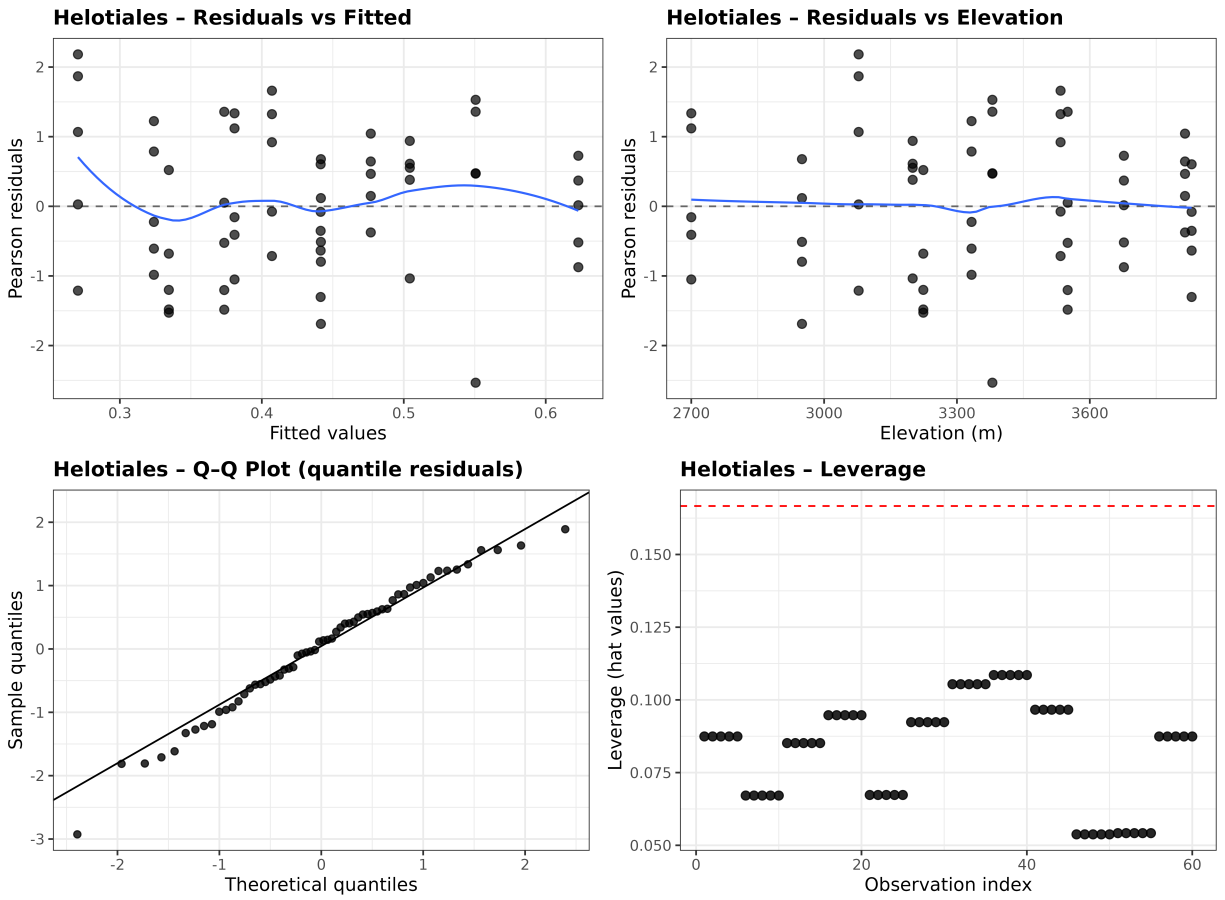


Figure S4: Diagnostic plots for the beta-regression model of Helotiales. *Code: see “Beta-regression models” below.*

Sebacinales model diagnostics

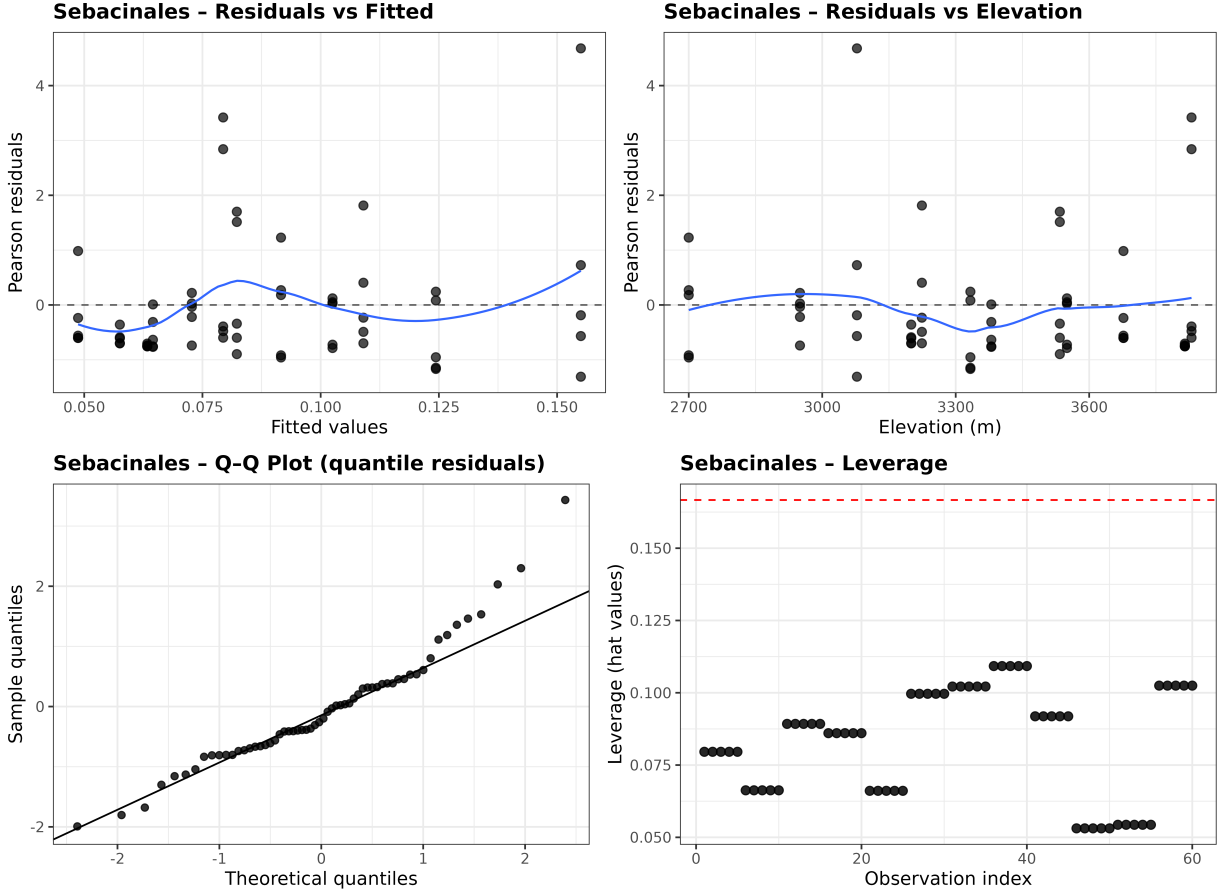


Figure S5: Diagnostic plots for the beta-regression model of Sebacinales. *Code: see “Beta-regression models” below.*

6.2 Figure 4: Elevational trends and genus habitat responses

Four-panel composite figure combining order-level elevational trends (beta-regression for the two most abundant and responsive orders) with genus-level habitat responses from per-genus GLMMs. Panels A and B show the fitted across-site beta-regression relationships for *Helotiales* and *Sebacinales* with raw data coloured by site. Panel C presents the genus-level GLMM heatmap of fold changes in subpáramo and páramo relative to forest. Panel D splits the *Helotiales* trend into its genus-assigned and genus-unassigned fractions.

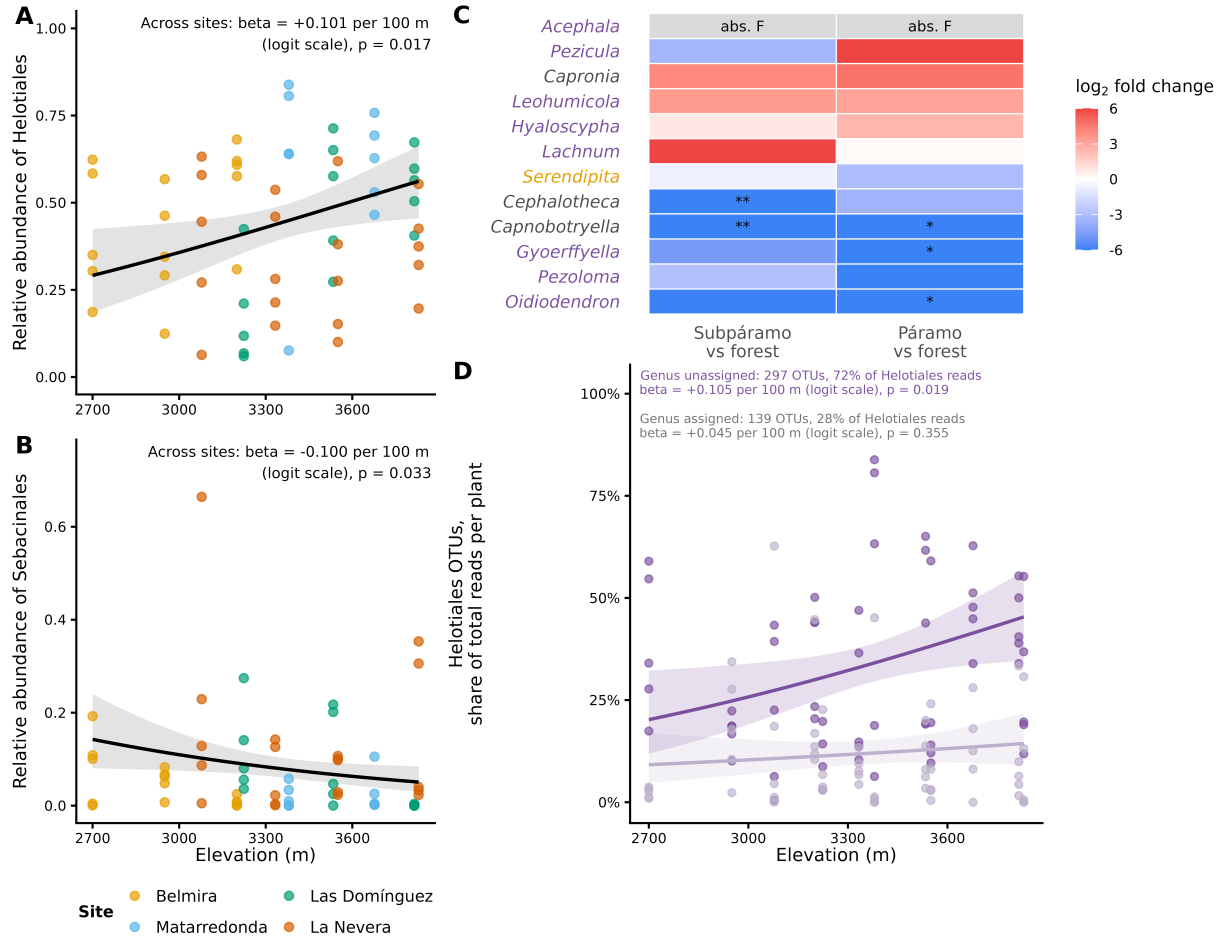


Figure 4: Elevational trends and genus-level habitat responses. Panel A: Helotiales relative abundance per plant versus elevation (logit-scale $\beta = +0.101$ per 100 m, $p = 0.017$; about +2.5 percentage points per 100 m at the mean proportion). Panel B: Sebaciniales relative abundance per plant versus elevation ($\beta = -0.100$ per 100 m, $p = 0.033$). Lines show the across-site fit of the additive model. Panel C: GLMM log fold changes for subpáramo vs forest and páramo vs forest for 12 of the 56 modelled genera, selected as (i) Helotiales or Sebaciniales genera with at least 40,000 reads, (ii) any genus with a resolved contrast against forest (BH $q < 0.05$), whatever its order or abundance, and (iii) *Capronia*, retained from the published figure as its most abundant Chaetothyriales genus. Criteria (i) and (ii) are independent of the estimated effect sizes; the complete 56-genus table is provided as `genus_glm_results.csv`. Stars mark BH-adjusted q-values (* $q < 0.05$, ** $q < 0.01$), and “abs.” marks the habitat where a genus was not detected (F = forest, S = subpáramo, P = páramo). Panel D: the Helotiales trend in Panel A split into genus-assigned and genus-unassigned OTUs, each fitted with the same additive beta regression (Table `helotiales_fraction_check.csv`). The increase is carried by the genus-unassigned fraction ($\beta = +0.105$ per 100 m, $p = 0.019$; 72% of Helotiales reads), which is about +2.3 percentage points per 100 m at its mean proportion, while genus-assigned OTUs show no trend ($\beta = +0.045$, $p = 0.355$). Coefficients in Panels A, B and D are all on the logit scale.

6.3 Genus-level GLMMs: habitat responses

We modelled genus-level habitat responses with a negative binomial generalized linear mixed model per genus (glmmTMB), replacing the earlier GLLVM, whose genus-level coefficients had diverged because many OTUs never occurred in the forest reference habitat. Reads were pooled to genus per plant (*Meliniomyces* merged into *Hyaloscypha*), and genera present in 10 plants with 1,000 reads were retained (56 genera). Each model was `count ~ habitat + site + offset(log(library size)) + (1 | location): site` as a

genus-specific fixed effect accounts for the partial confounding of habitat and site, and the location random effect makes the 12 sampling locations the replicates for habitat. Contrasts against forest are reported as fold changes with 95% confidence intervals and Benjamini–Hochberg q-values across genera; genera absent from a habitat are flagged rather than given a fold change.

Four genera had a significant contrast against forest, and every one is a decline (*Capnobotryella*, *Cephalotheca*, *Oidiodendron*, *Gyoerffyyella*); no genus was significantly more abundant in subpáramo or páramo than in forest. *Acephala* had the strongest habitat support of any genus (omnibus $q = 0.029$) but occurs only at two Las Domínguez locations and was never detected in forest. The order-level Helotiales increase (Section 6.1) is carried by genus-unassigned OTUs (72% of Helotiales reads; +11.1% in the odds per 100 m, $p = 0.019$), whereas genus-assigned Helotiales show no trend ($p = 0.36$).

Figure 4C shows 12 of the 56 modelled genera. Rows are selected before any effect size is examined: Helotiales and Sebaciniales genera above an abundance floor of 40,000 reads, together with every genus whose contrast against forest is resolved at BH $q < 0.05$ regardless of order or abundance, so that each significant result appears in the figure. *Capronia* is retained as a single named exception, being the most abundant Chaetothyriales genus of the published figure; no abundance threshold admits it without also admitting four more abundant genera that have no habitat support. This replaces the earlier selection, which carried all 14 genera of the published GLLVM figure whether or not they were supported.

The 12 genera of Figure 4C are previewed below, in the order they appear in the panel. The complete 56-genus table is provided as Data S2 (`Data_files/Data_S2_genus_GLMM_table.csv`, identical to `tables/genus_glmm_results.csv`).

Table 18: Data S2 (preview). The 12 genera shown in Figure 4C, in the same top-to-bottom order as the panel, from the per-genus negative binomial GLMMs. Plants F/S/P = number of plants in which the genus was detected in forest, subpáramo and páramo; S/F and P/F = fold change in subpáramo and páramo relative to forest, with Benjamini–Hochberg q-values across genera. A genus never detected in forest has no estimable fold change against it (shown as NA here and as “abs. F” in Figure 4C), and a contrast whose fit returned no usable standard error has a fold change but no confidence limits or q-value (NA). The complete 56-genus table is provided as Data S2.

Genus	Order	Plants F	Plants S	Plants P	S/F ratio	S/F q	P/F ratio	P/F q
Acephala	Helotiales	0	5	5	NA	NA	NA	NA
Pezicula	Helotiales	12	4	8	9.88e-02	0.76200	8.59e+02	0.1460
Capronia	Chaetothyriales	10	6	5	1.62e+01	0.29100	2.63e+01	0.4110
Leohumicola	Helotiales	7	6	2	1.11e+01	0.76200	8.98e+00	0.5490
Hyaloscypha	Helotiales	8	9	6	1.78e+00	0.86100	5.57e+00	0.5340
Lachnum	Helotiales	6	7	3	1.02e+03	0.05920	1.16e+00	0.9880
Serendipita	Sebaciniales	22	14	10	6.22e-01	0.85500	1.19e-01	0.1840
Cephalotheca	Cephalothecales	11	1	5	2.66e-05	0.00943	8.98e-02	0.5490
Capnobotryella	Mycosphaerellales	9	2	1	4.51e-05	0.00943	3.12e-05	0.0141
Gyoerffyyella	Helotiales	14	4	4	3.82e-02	0.27800	4.14e-03	0.0131
Pezoloma	Helotiales	5	5	1	1.33e-01	NA	2.03e-03	NA
Oidiodendron	Helotiales	17	8	1	1.29e-02	0.17400	6.33e-05	0.0141

Section 7: Functional guilds

Functional guilds from root samples were assigned using FUNGuild based on the curated taxonomic strings exported from the phyloseq object. OTUs with a valid confidence ranking in the FUNGuild output were retained for downstream summaries. Guild labels were additionally collapsed into a primary-guild hierarchy (Ericoid mycorrhizal > other mycorrhizal > endophyte > pathotroph > saprotroph > other) to simplify interpretation. Figure S6 summarizes mean guild-associated read abundance per sample across habitats.

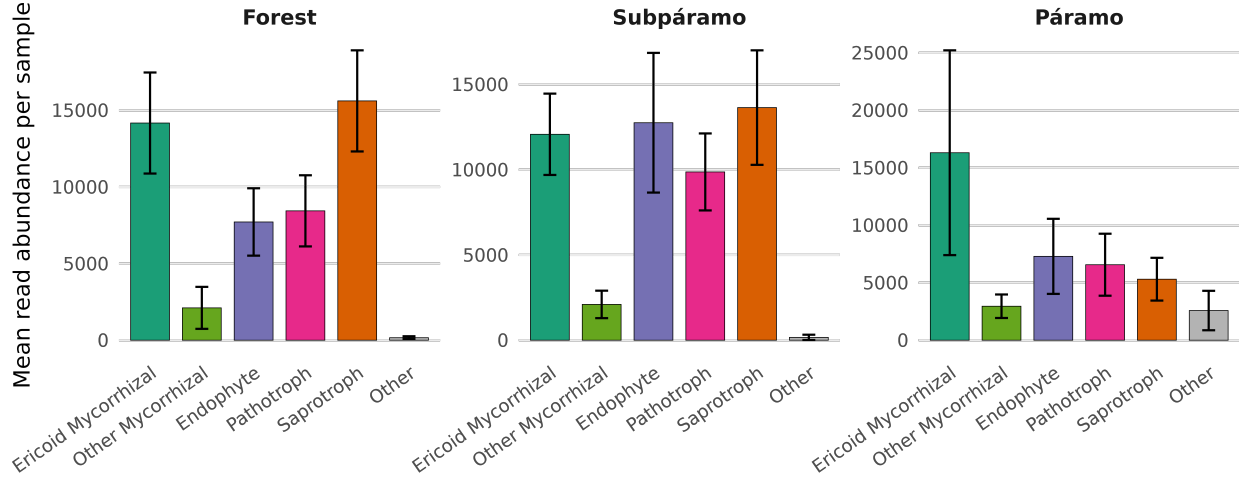


Figure S6: Mean relative read abundance per sample of fungal functional guilds across forest, subpáramo, and páramo habitats, based on FUNGuild annotations. Only root samples. *Code: see “FUNGuild functional annotation” below.*

Section 8: Phylogenetic community structure

Phylogenetic community structure was quantified using Net Relatedness Index (NRI; based on MPD) and Nearest Taxon Index (NTI; based on MNTD), calculated under a taxa-label null model with 999 randomizations using both abundance-weighted (aw) and presence-absence (pa) versions. Indices were computed per plant ($n = 60$). Habitat-level summaries are reported as median (IQR) (Table S19). Habitat and site effects on abundance-weighted metrics were evaluated with linear models, each adjusted for the other (habitat + site; Table S20): neither index differed among habitats (NRI $p = 0.48$; NTI $p = 0.43$), whereas NRI differed among sites ($F_{3,54} = 3.29$, $p = 0.027$) and NTI did not ($p = 0.19$).

Table 19: Table S19. Median (IQR) of per-plant NRI and NTI across habitats, calculated using abundance-weighted (aw) and presence-absence (pa) metrics. Positive values indicate phylogenetic clustering relative to null expectations.

habitat	n	NRI_aw	NTI_aw	NRI_pa	NTI_pa
forest	25	1.057 (3.01)	3.111 (1.96)	0.631 (2.16)	5.750 (2.32)
subparamo	20	1.698 (1.69)	2.779 (1.40)	0.385 (1.50)	4.928 (2.54)
paramo	15	1.419 (1.80)	3.253 (1.66)	0.838 (2.83)	4.804 (1.98)

Table 20: Table S20. Tests for linear models of per-plant abundance-weighted NTI and NRI: habitat and site each adjusted for the other (habitat + site), and the habitat $\times$ site interaction by model comparison.

response	test	df1	df2	F	p
NRI_aw	habitat adjusted for site	2	54	0.7504	0.4770
NRI_aw	site adjusted for habitat	3	54	3.2925	0.0273
NRI_aw	habitat x site interaction	4	50	2.5154	0.0530
NTI_aw	habitat adjusted for site	2	54	0.8615	0.4283
NTI_aw	site adjusted for habitat	3	54	1.6435	0.1901
NTI_aw	habitat x site interaction	4	50	1.9889	0.1105

Section 9: Soil physicochemical context

To provide environmental context for the elevational gradients sampled in this study, we quantified bulk soil physicochemical properties from one composite topsoil sample (0–15 cm) collected at each sampling location ($n = 12$; one per location). Because soil chemistry was measured once per location, these data are interpreted descriptively and were not used for formal statistical inference among habitats or sites.

The full soil dataset, including all measured variables is archived in the repository as “Data_S3_soil.csv”.

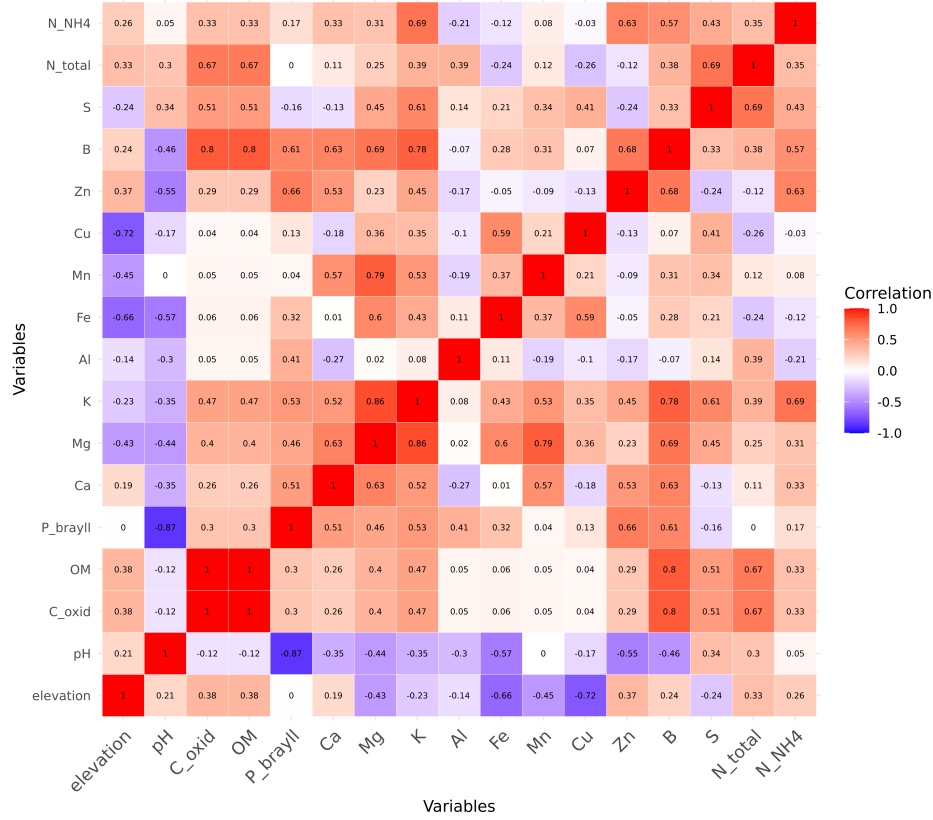


Figure S7: Correlation heatmap of bulk soil physicochemical variables measured at each sampling location ($n = 12$). Colors indicate Pearson correlation among soil variables and elevation.